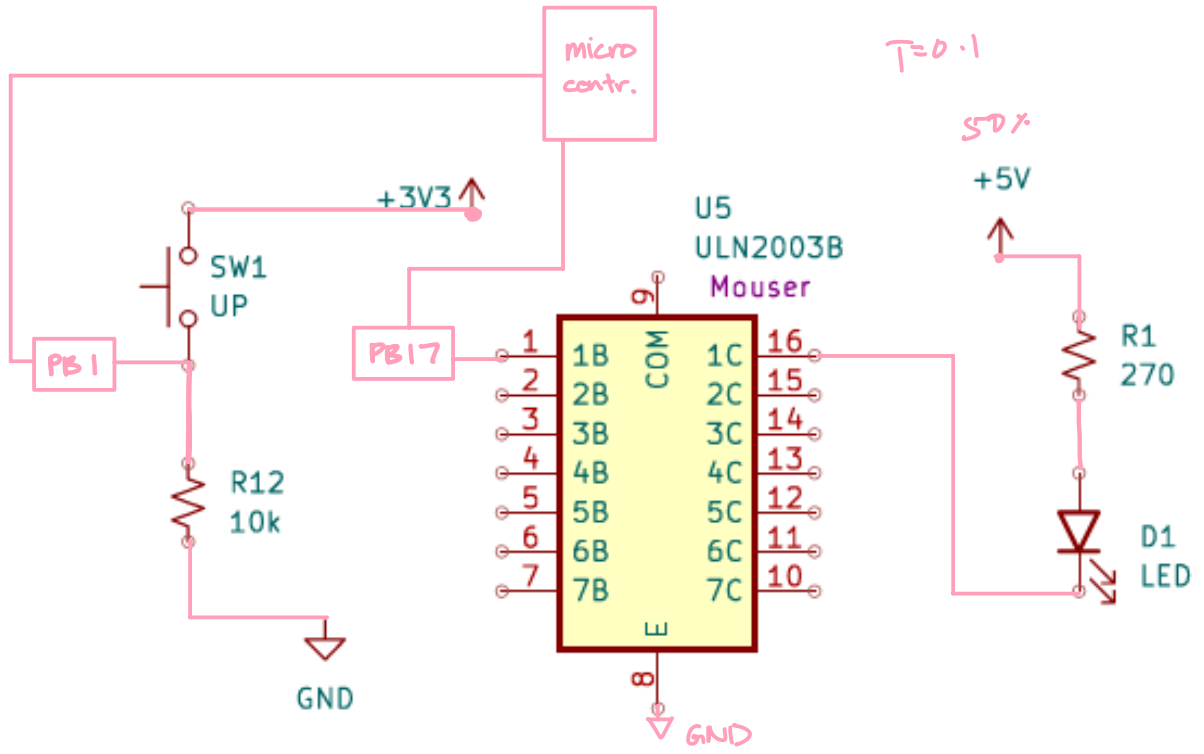
0) Melinda Dai

Dr. Holt

mmd3728

1)



$Freq = \frac{1}{T} = 10MHz$

$T = 0.1\mu s$

50%

$\frac{time \times 10MHz}{cycle} = \#cycles$

$\frac{0.1\mu s \times 8MHz}{4cycles} = 2000000$

2) $V_{supply} = I_d R + V_d + V_{CE}$

$I_d = \frac{V_{supply} - V_{CE} - V_d}{R}$

mA $I_d = \frac{5 - 0.4 - V_d}{270} \cdot 1000$

$I_d = 3.7037(4.6 - V_d)$

$I_d = 20V_d - 32$

$3.7037(4.6 - V_d) = 20V_d - 32$

$V_d \approx 2.069\,V \qquad I_d = 20(2.069) - 32 \approx 9.38\,mA$

check: on graph, V_f @ ($I_F \approx 9.4mA$) $\approx 1.9V$

$V_F \approx 2.069V$

$I_F \approx 9.38\,mA$

3)

Parameter	Value	Units	Conditions
Input Voltage on PB0	0 V	volts	With switch not pressed (measured with voltmeter)
Resistor current	0 mA	mA	With switch not pressed (measured with ~~ammeter~~) voltmeter
Input Voltage on PB0	3.18 V	volts	With switch pressed (measured with voltmeter)
Resistor current	0.318 mA	mA	With switch pressed (measured with ~~ammeter~~) voltmeter

Table 2.1. Switch measurements.

4)

Parameter	Value	Units	Conditions
LED Voltage, V_d	0V	volts	With LED off (measured with voltmeter)
LED Current, I_d	0 mA	mA	With LED off (measured with ~~ammeter~~) voltmeter
LED Voltage, V_d	2V	volts	With LED on (measured with voltmeter)
LED Current, I_d	8.63 mA	mA	With LED on (measured with ~~ammeter~~) voltmeter
LED Power, $V_d * I_d$	17.259 mW	mW	With LED on (multiply two measurements)

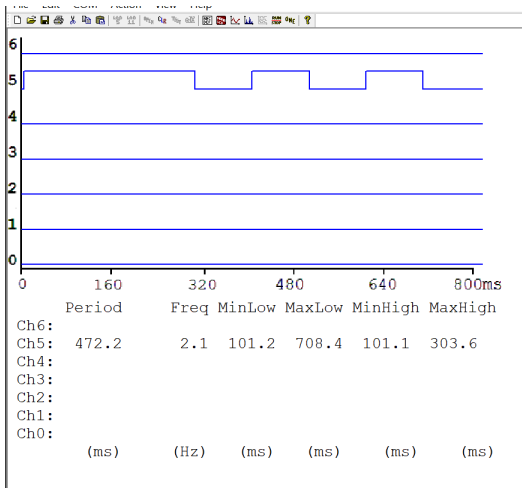
Table 2.2. LED measurements.

$8.63 = 20V_d - 32$

$V_d = 1.6V \neq 2.069V$

→ Equation is a simplified version of the curve in the graph

5)



$\frac{1}{20Hz} = T$

90% // 20 Hz T

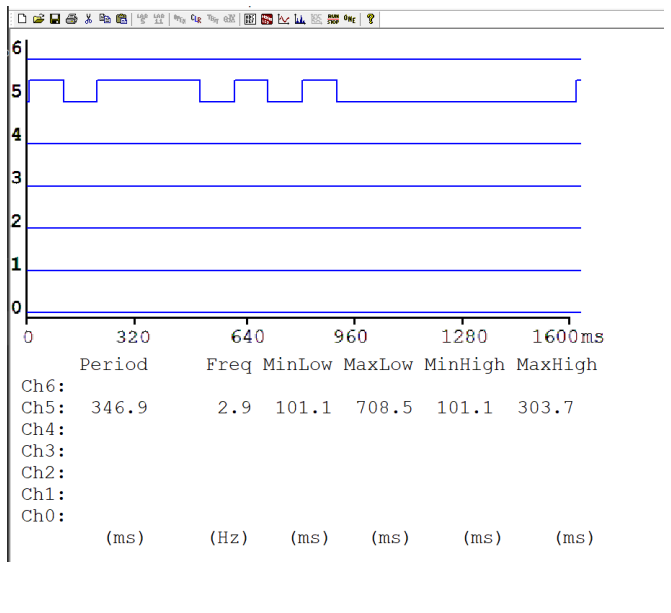
$\frac{H}{H+L} \times 100 = 90\%$

on for $0.9 \times T$ $0.9 \times \frac{1}{20Hz}$ = time

off or 0.1

$\frac{0.9 \times T \times 20Hz}{4} = \#cycles$

6)



7)

```
ECE319K Lab 2H, Spring 2026
Student EID= MMD3728
Send the letter: Q, Wave= D.D.d.D Yes, Score=5
Send the letter: E, Wave= d Yes, Score=10
Send the letter: F, Wave= d.d.D.d Yes, Score=15
Send the letter: M, Wave= D.D Yes, Score=20
Send the letter: V, Wave= d.d.d.D Perfect,  Score=25
End of Lab 2H, Spring 2026
```

100ns

$$\frac{0.1 \times 8\text{mHz}}{4} = 20000$$

$10 = \text{mHz}$

time $\times$ 10mHz

total

delay $\rightarrow$ $\frac{0.5 \times 10\ \text{mHz}}{4} = \#\ \text{loops}$

ON $0.5 \times T$ = time on

OFF $0.5 \times T$ = time off

time on $\times$ 10MHz = # cycles on

time off $\times$ 10 MHz = # cycles of